Diskrete Mathematik – Woche 7

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https://github.com/tschai-yim/eth-resources
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Functions

Problem 1 ⋆

Let $A = \{1, 2, 3\}$ and $B = \{x, y\}$.

1. Which of the following relations from A to B are functions $A \to B$? Which are partial functions (Def. 3.35)?

$$\rho_1 = \{(1, x), (2, y), (3, x)\}, \quad \rho_2 = \{(1, x), (2, y)\}, \quad \rho_3 = \{(1, x), (1, y), (2, y), (3, y)\}, \quad \rho_4 = \varnothing$$

2. Determine $|B^A|$, the number of partial functions $A \to B$, and the number of relations from A to B.
3. Let $C \neq \varnothing$ be a set. Determine the sets $C^\varnothing$, $\varnothing^C$ and $\varnothing^\varnothing$ (Def. 3.34).
4. Which of the following relations on $\mathbb{R}$ are functions $\mathbb{R} \to \mathbb{R}$?

$$\sigma_1 = \{(u, v) \mid v = u^2\}, \quad \sigma_2 = \{(u, v) \mid u = v^2\}, \quad \sigma_3 = \{(u, v) \mid u^2 + v^2 = 1\}$$

Problem 2 ⋆ ⋆

Let $f : A \to B$ be a function and $S, S' \subseteq A$. Prove or disprove:

1. $f(S \cup S') = f(S) \cup f(S')$
2. $f(S \cap S') = f(S) \cap f(S')$
3. ($\star\star\star$) Prove: f is injective $\Leftrightarrow f(S \cap S') = f(S) \cap f(S')$ for all $S, S' \subseteq A$.

Problem 3 ⋆ ⋆

Let $f : A \to B$ and $g : B \to C$ be functions.

1. Express the function $g \circ f$ (Def. 3.41) as a composition of relations in the sense of Def. 3.12.
2. Prove: if f and g are injective, then $g \circ f$ is injective. If f and g are surjective, then $g \circ f$ is surjective.
   (The first statement is used without proof in Lemma 3.15 (ii).)
3. Prove: if $g \circ f$ is injective, then f is injective. If $g \circ f$ is surjective, then g is surjective.
4. Find A, B, C, f, g such that $g \circ f$ is bijective, but f is not surjective and g is not injective.

Equinumerosity

Problem 4 ⋆ ⋆

1. Let X be an infinite set, and let $a_0, a_1, a_2, \ldots \in X$ be pairwise distinct (an infinite sequence in X without repetitions). We show that removing a_0 does not change the "size" of X, i.e. $X \sim X \setminus \{a_0\}$.
   (a) Define a function $h : X \to X \setminus \{a_0\}$ that shifts the sequence by one position ($a_0 \mapsto a_1$, $a_1 \mapsto a_2$, $a_2 \mapsto a_3$, ...) and maps every other element of X to itself. Write h down as a case distinction.
   (b) Prove that h indeed maps into $X \setminus \{a_0\}$, that h is injective, and that h is surjective.
2. Use 1. to show $\mathbb{N} \sim \mathbb{N} \setminus \{0\}$.
3. Use 1. (twice) and Lemma 3.15 to show $[0, 1] \sim (0, 1)$.

Countable and Uncountable Sets

Problem 5 ⋆ ⋆

Decide whether each set is countable or uncountable. Justify with results from the script.

1. $\mathbb{Z} \times \mathbb{Q}$
2. $\mathcal{F} = \{S \subseteq \mathbb{N} \mid S \text{ is finite}\}$
3. $\mathcal{P}(\mathbb{N})$

Problem 6 ⋆ ⋆

Let $\mathcal{I} = \{S \subseteq \mathbb{N} \mid S \text{ is infinite}\}$.

1. Prove that $\mathcal{I}$ is uncountable by giving an explicit injection $\varphi : \{0,1\}^\infty \to \mathcal{I}$. Prove that $\varphi(s) \in \mathcal{I}$ for all s and that φ is injective.
2. The map $\chi : s \mapsto \{i \in \mathbb{N} \mid s_i = 1\}$ from Problem 5 is injective, but it is not a function $\{0,1\}^\infty \to \mathcal{I}$. Why not?

Problem 7 ⋆ ⋆ ⋆

1. Let A be countable and $f : A \to B$ surjective. Prove that B is countable.
   (Hint: use an injection $h : A \to \mathbb{N}$ to pick, for each $b \in B$, one element of $f^{-1}(\{b\})$.)
2. Prove or disprove: if A is uncountable and $f : A \to B$ is surjective, then B is uncountable.

Problem 8 ⋆ ⋆ ⋆

Let A be an arbitrary set.

1. Show $A \preceq \mathcal{P}(A)$.
2. Let $f : A \to \mathcal{P}(A)$ be any function and $D = \{a \in A \mid a \notin f(a)\}$. Prove that $D \notin f(A)$. Conclude $A \not\approx \mathcal{P}(A)$.
3. Deduce that $\mathcal{P}(\mathbb{N})$ is uncountable, without using Theorem 3.23.
4. Let $A = \mathbb{N}$ and identify each subset $S \subseteq \mathbb{N}$ with the sequence $s \in \{0,1\}^\infty$ where $s_i = 1 \Leftrightarrow i \in S$. Which object from the proof of Theorem 3.23 corresponds to D?

Problem 9 ⋆ ⋆ (extra)

1. Let A, B be finite with $|A| = |B|$ and $f : A \to B$. Prove: f injective $\Leftrightarrow$ f surjective.
2. Show that both directions fail for $A = B = \mathbb{N}$.

Problem 10 ⋆ ⋆ ⋆ (extra)

Let $\mathcal{J}$ be a set of pairwise disjoint open intervals $(a, b) \subseteq \mathbb{R}$ with $a < b$.

1. Prove that $\mathcal{J}$ is countable. (You may use: between any two reals $a < b$ there is a rational number.)
2. Does 1. still hold for pairwise disjoint closed intervals $[a, b]$ with $a \leq b$?